Distribution	Density	Distribution	Density
Exponential	$\lambda e^{-\lambda x}$	Uniform	$\frac{1}{b-a}\ \frac{1}{b-a}\mathbf{1}_{[a,b]}(x),\ a<b$
Laplace	$\frac{1}{2b}e^{-\frac{\lvert x-\mu\rvert}{b}}$	Gaussian	$\frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
Cauchy	$\frac{1}{\pi\gamma\left[1+\left(\frac{x-x_0}{\gamma}\right)^2\right]}$	Beta	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}$
Student-T	$c\left(1+\frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$	LogNormal	$\frac{1}{x\sigma\sqrt{2\pi}}e^{-\frac{(\ln x-\mu)^2}{2\sigma^2}}$
Chi-squared	$\frac{1}{2^{k/2}\Gamma(k/2)}x^{k/2-1}e^{-x/2}$	Gamma	$\frac{\beta^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\beta x}$
Logistic	$\frac{e^{-(x-\mu)/s}}{s(1+e^{-(x-\mu)/s})^2}$	Pareto	$\frac{\alpha x_m^\alpha}{x^{\alpha+1}}$
Rayleigh	$\frac{x}{\sigma^2}e^{-x^2/(2\sigma^2)}$	Weibull	$\frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1}e^{-(x/\lambda)^k}$
		Gumbel-1	$\frac{1}{\beta}e^{-(z+e^{-z})},\quad z=\frac{x-\mu}{\beta}$
		Gumbel-2	$\frac{\alpha\beta x^{-\alpha-1}e^{-(x/\beta)^{-\alpha}}}{\beta}$
		Exponentiated Power	$k\lambda x^{k-1}e^{-(x/\lambda)^k}$
Discrete Distributions			
Poisson	$\frac{\lambda^k e^{-\lambda}}{k!}$	Geometric	$(1-p)^{k-1}p$
Binomial	$\binom{n}{k}p^k(1-p)^{n-k}$	Negative Binomial	$\binom{k+r-1}{k}(1-p)^r p^k$
Hypergeometric	$\frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$	Bernoulli	$p^k(1-p)^{1-k}$
Discrete Uniform	$\frac{1}{n}$	Multinomial	$\frac{n!}{x_1!\cdots x_k!}p_1^{x_1}\cdots p_k^{x_k}$

- $e_1, e_2 \in \mathbb{R}$ and $e_1 > e_2 e_1 \geq e_2$: then $\mathcal{D}[\![\texttt{uniform}(\ldots) : \texttt{float}[B;\top]]\!] = \texttt{diverge}$, which is well-typed at any type by Diverge.
- $B = \top$ and the literal-invalid condition above does not hold: then $\mathcal{D}[\![\texttt{float}[B;\top]]\!] = \texttt{float}$, and

  $\mathcal{D}[\![\texttt{uniform}(e_1, e_2) : \texttt{float}[B;\top]]\!] = \texttt{uniform}(\mathcal{D}[\![e_1 : \texttt{float}[B_1;V_1]]\!], \mathcal{D}[\![e_2 : \texttt{float}[B_2;V_2]]\!])$.

  Since recovers(B_1, V_1) and recovers(B_2, V_2) are preserved under discretization, the result is well-typed at $\texttt{float}$ by Continuous-2.
- $B \neq \top$and , $e_1, e_2 \in \mathbb{R}$, and $e_1 < e_2$: then $\mathcal{D}[\![\texttt{float}[B;\top]]\!] = \texttt{fin}(|B|+1)$, and

  $$\mathcal{D}[\![\texttt{uniform}(e_1, e_2) : \texttt{float}[B;\top]]\!] = \texttt{discrete}(p_0, \ldots, p_{|B|})$$

  where $p_k = \text{CDF}(b_{k+1}) - \text{CDF}(b_k)$. Each $p_k = \Pr[X \in \text{Intervals}(B)_k]$ for $X \sim \text{Uniform}(e_1, e_2)$. Hence the distribution is atomless, each $p_k \in [0,1]$, and $\sum_k p_k = 1$, so this is well-typed at $\texttt{fin}(|B|+1)$ by Discrete.
- $B \neq \top$, $B_1 \neq \top$, $B_2 \neq \top$, and e_1 or e_2 is a variableat least one of e_1, e_2 is nonliteral: the algorithm generates conditionals over all pairs (v_i, w_j), each branch dispatching to a constant parameter in $\texttt{uniform}$. By the previous sub-case, each valid branch $\mathcal{D}[\![\texttt{uniform}(v_i, w_j) : \texttt{float}[B;\top]]\!]$ is well-typed at $\texttt{fin}(|B|+1) = \mathcal{D}[\![\texttt{float}[B;\top]]\!]$, while equality or reversed bounds use the malformed branch. The conditional structure is well-typed by If (as in the $\texttt{if}$ case above), so the full expression is well-typed at $\mathcal{D}[\![\texttt{float}[B;\top]]\!]$.

Case e is any other language construct (functions, pairs, lists, *etc.*). By definition of $\mathcal{D}[\![\cdot]\!]$, the discretization recurses structurally over e and τ. That is, $\mathcal{D}[\![e : \tau]\!]$ is formed by applying $\mathcal{D}[\![\cdot]\!]$ to each subexpression and subtype of $e : \tau$, and $\mathcal{D}[\![\tau]\!]$ is formed by applying $\mathcal{D}[\![\cdot]\!]$ to each component type of τ. The result then follows immediately by applying the inductive hypothesis to each subterm, analogous to the above cases.

In every case, the typing derivation is preserved under $\mathcal{D}[\![\cdot]\!]$. Therefore,

$$\mathcal{D}[\![\Gamma]\!] \vdash \mathcal{D}[\![e : \tau]\!] : \mathcal{D}[\![\tau]\!].$$ □

$\mathbf{e} : \boldsymbol{\tau}$	constraints	$[\![\boldsymbol{e} : \boldsymbol{\tau}, \mathcal{D}[\![\boldsymbol{e}]\!] : \mathcal{D}[\![\boldsymbol{\tau}]\!]]\!]$
$e_1 : \tau_1$ && $e_2 : \tau_2 : \mathbf{bool}$ $\tau_1 = \mathbf{bool}, \tau_2 = \mathbf{bool}$	$e_1 = b_1 = \text{true},\ e_2 = b_2 = \text{true}$ $e_1 = b_1,\ e_2 = b_2$ $e_1 = b_1,\ e_2 \neq b_2$ o.w.	$\delta_{(\text{true}:\mathbf{bool},\text{true}:\mathbf{bool})}$ $\delta_{(\text{false}:\mathbf{bool},\text{false}:\mathbf{bool})}$ $[\![e_2 : \tau_2, \mathcal{D}[\![e_2 : \tau_2]\!] : \mathcal{D}[\![\tau_2]\!]]\!]$ $\gg\!= \lambda(g,g').\delta_{(b_1:\tau_1 \&\& g:\mathbf{bool},\ b_1:\mathcal{D}[\![\tau_1]\!] \&\& g':\mathcal{D}[\![\mathbf{bool}]\!])}$ $[\![e_1 : \tau_1, \mathcal{D}[\![e_1 : \tau_1]\!] : \mathcal{D}[\![\tau_1]\!]]\!]$ $\gg\!= \lambda(g,g').\delta_{(g \&\& e_2:\tau_2:\mathbf{bool},\ g' \&\& \mathcal{D}[\![e_2:\tau_2]\!]:\mathcal{D}[\![\tau_2]\!]:\mathcal{D}[\![\mathbf{bool}]\!])}$
$e_1 : \tau_1$ || $e_2 : \tau_2 : \mathbf{bool}$ $\tau_1 = \mathbf{bool}, \tau_2 = \mathbf{bool}$	$e_1 = b_1 = \text{false},\ e_2 = b_2 = \text{false}$ $e_1 = b_1,\ e_2 = b_2$ $e_1 = b_1,\ e_2 \neq b_2$ o.w.	$\delta_{(\text{false}:\mathbf{bool},\text{false}:\mathbf{bool})}$ $\delta_{(\text{true}:\mathbf{bool},\text{true}:\mathbf{bool})}$ $[\![e_2 : \tau_2, \mathcal{D}[\![e_2 : \tau_2]\!] : \mathcal{D}[\![\tau_2]\!]]\!]$ $\gg\!= \lambda(g,g').\delta_{(b_1:\tau_1 \,\|\| g:\mathbf{bool},\ b_1:\mathcal{D}[\![\tau_1]\!] \,\|\| g':\mathcal{D}[\![\mathbf{bool}]\!])}$ $[\![e_1 : \tau_1, \mathcal{D}[\![e_1 : \tau_1]\!] : \mathcal{D}[\![\tau_1]\!]]\!]$ $\gg\!= \lambda(g,g').\delta_{(g \,\|\| e_2:\tau_2:\mathbf{bool},\ g' \,\|\, \mathcal{D}[\![e_2:\tau_2]\!]:\mathcal{D}[\![\tau_2]\!]:\mathcal{D}[\![\mathbf{bool}]\!])}$
not $e : \tau : \mathbf{bool}$ $\tau = \mathbf{bool}$	$e = \text{true}$ $e = \text{false}$ o.w.	$\delta_{(\text{false}:\mathbf{bool},\text{false}:\mathbf{bool})}$ $\delta_{(\text{true}:\mathbf{bool},\text{true}:\mathbf{bool})}$ $[\![e : \tau, \mathcal{D}[\![e : \tau]\!] : \mathcal{D}[\![\tau]\!]]\!]$ $\gg\!= \lambda(g,g').\delta_{(\text{not } g:\mathbf{bool},\ \text{not } g':\mathcal{D}[\![\mathbf{bool}]\!])}$
$(e_1 : \tau_1, e_2 : \tau_2) : \tau_1 * \tau_2$	$e_1 = v_1,\ e_2 = v_2$ $e_1 = v_1,\ e_2 \neq v_2$ o.w.	$\delta_{((v_1,v_2):\tau_1*\tau_2,\ (\mathcal{D}[\![v_1]\!]:\mathcal{D}[\![\tau_1]\!],\mathcal{D}[\![v_2]\!]:\mathcal{D}[\![\tau_2]\!]):\mathcal{D}[\![\tau_1]\!]*\mathcal{D}[\![\tau_2]\!])}$ $[\![e_2 : \tau_2, \mathcal{D}[\![e_2 : \tau_2]\!] : \mathcal{D}[\![\tau_2]\!]]\!]$ $\gg\!= \lambda(g,g').\delta_{((v_1:\tau_1,g:\tau_2):\tau_1*\tau_2,\ (\mathcal{D}[\![v_1]\!]:\mathcal{D}[\![\tau_1]\!],g':\mathcal{D}[\![\tau_2]\!]):\mathcal{D}[\![\tau_1]\!]*\mathcal{D}[\![\tau_2]\!])}$ $[\![e_1 : \tau_1, \mathcal{D}[\![e_1 : \tau_1]\!] : \mathcal{D}[\![\tau_1]\!]]\!]$ $\gg\!= \lambda(g,g').\delta_{((g,e_2:\tau_2):\tau_1*\tau_2,\ (g',\mathcal{D}[\![e_2:\tau_2]\!]):\mathcal{D}[\![\tau_1]\!]*\mathcal{D}[\![\tau_2]\!])}$
$\mathbf{fst}\ (e : \tau_1 * \tau_2) : \tau_1$	$e = (v_1, v_2)$ o.w.	$\delta_{(v_1:\tau_1,\ \mathcal{D}[\![v_1:\tau_1]\!]:\mathcal{D}[\![\tau_1]\!])}$ $[\![e : \tau_1 * \tau_2, \mathcal{D}[\![e]\!] : \mathcal{D}[\![\tau_1 * \tau_2]\!]]\!]$ $\gg\!= \lambda(g,g').\delta_{(\mathbf{fst}\ (g:\tau_1*\tau_2):\tau_1,\ \mathbf{fst}\ (g':\mathcal{D}[\![\tau_1*\tau_2]\!]):\mathcal{D}[\![\tau_1]\!])}$
$\mathbf{snd}\ (e : \tau_1 * \tau_2) : \tau_2$	$e = (v_1, v_2)$ o.w.	$\delta_{(v_2:\tau_2,\ \mathcal{D}[\![v_2:\tau_2]\!]:\mathcal{D}[\![\tau_2]\!])}$ $[\![e : \tau_1 * \tau_2, \mathcal{D}[\![e]\!] : \mathcal{D}[\![\tau_1 * \tau_2]\!]]\!]$ $\gg\!= \lambda(g,g').\delta_{(\mathbf{snd}\ (g:\tau_1*\tau_2):\tau_2,\ \mathbf{snd}\ (g':\mathcal{D}[\![\tau_1*\tau_2]\!]):\mathcal{D}[\![\tau_2]\!])}$
$\mathbf{fun}\ x \to e : \tau_1 \to \tau_2$		$\delta_{(\mathbf{fun}\ x \to e:\tau_1\to\tau_2,\ \mathbf{fun}\ x \to \mathcal{D}[\![e]\!]:\mathcal{D}[\![\tau_1]\!]\to\mathcal{D}[\![\tau_2]\!])}$
$\mathbf{fix}\ g\ x := e : \tau_1 \to \tau_2$		$\delta_{(\mathbf{fix}\ g\ x:=e:\tau_1\to\tau_2,\ \mathbf{fix}\ g\ x:=\mathcal{D}[\![e]\!]:\mathcal{D}[\![\tau_1]\!]\to\mathcal{D}[\![\tau_2]\!])}$
$(e_1 : \tau_1 \to \tau_2\ \ e_2 : \tau_1) : \tau_2$	$e_1 = \mathbf{fun}\ x \to e$ or $\mathbf{fix}\ f\ x := e$, $e_2 = v_2$ $e_1 = \mathbf{fun}\ x \to e$ or $\mathbf{fix}\ f\ x := e$, $e_2 \neq v_2$ o.w.	$\delta_{(e[v_2/x]:\tau_2,\ \mathcal{D}[\![e]\!][\mathcal{D}[\![v_2]\!]/x]:\mathcal{D}[\![\tau_2]\!])}$ $[\![e_2 : \tau_1, \mathcal{D}[\![e_2 : \tau_1]\!] : \mathcal{D}[\![\tau_1]\!]]\!]$ $\gg\!= \lambda(g,g').\delta_{(e_1:\tau_1\to\tau_2\ \ g:\tau_1:\tau_2,\ \mathcal{D}[\![e_1]\!]:\mathcal{D}[\![\tau_1]\!]\to\mathcal{D}[\![\tau_2]\!]\ \ g':\mathcal{D}[\![\tau_1]\!]:\mathcal{D}[\![\tau_2]\!])}$ $[\![e_1 : \tau_1 \to \tau_2, \mathcal{D}[\![e_1]\!] : \mathcal{D}[\![\tau_1 \to \tau_2]\!]]\!]$ $\gg\!= \lambda(g,g').\delta_{(g\ e_2:\tau_1:\tau_2,\ g'\ \ \mathcal{D}[\![e_2]\!]:\mathcal{D}[\![\tau_1]\!]:\mathcal{D}[\![\tau_2]\!])}$
nil : $\mathbf{list}(\tau)$		$\delta_{(\text{nil}:\mathbf{list}(\tau),\ \text{nil}:\mathbf{list}(\mathcal{D}[\![\tau]\!]))}$
$(e_1 : \tau :: e_2 : \mathbf{list}(\tau)) : \mathbf{list}(\tau)$	$e_1 = v_1,\ e_2 = v_2$ $e_1 = v_1,\ e_2 \neq v_2$ o.w.	$\delta_{(v_1::v_2:\mathbf{list}(\tau),\ \mathcal{D}[\![v_1]\!]::\mathcal{D}[\![v_2]\!]:\mathbf{list}(\mathcal{D}[\![\tau]\!]))}$ $[\![e_2 : \mathbf{list}(\tau), \mathcal{D}[\![e_2]\!] : \mathbf{list}(\mathcal{D}[\![\tau]\!])]\!]$ $\gg\!= \lambda(g,g').\delta_{(v_1::g:\mathbf{list}(\tau),\ \mathcal{D}[\![v_1]\!]::g':\mathbf{list}(\mathcal{D}[\![\tau]\!]))}$ $[\![e_1 : \tau, \mathcal{D}[\![e_1]\!] : \mathcal{D}[\![\tau]\!]]\!]$ $\gg\!= \lambda(g,g').\delta_{(g::e_2:\mathbf{list}(\tau):\mathbf{list}(\tau),\ g'::\mathcal{D}[\![e_2]\!]:\mathbf{list}(\mathcal{D}[\![\tau]\!]):\mathbf{list}(\mathcal{D}[\![\tau]\!]))}$
$(\mathbf{match}\ e : \mathbf{list}(\tau_1)\ \mathbf{with}$ $\quad \mid \text{nil} \to e_1 : \tau_2$ $\quad \mid h :: t \to e_2 : \tau_2$ $\quad \mathbf{end}) : \tau_2$	$e = \text{nil}$ $e = v_h :: v_t$ o.w.	$\delta_{(e_1:\tau_2,\ \mathcal{D}[\![e_1:\tau_2]\!]:\mathcal{D}[\![\tau_2]\!])}$ $\delta_{(e_2[v_h/h,v_t/t]:\tau_2,\ \mathcal{D}[\![e_2:\tau_2]\!][\mathcal{D}[\![v_h]\!]/h,\mathcal{D}[\![v_t]\!]/t]:\mathcal{D}[\![\tau_2]\!])}$ $[\![e : \mathbf{list}(\tau_1), \mathcal{D}[\![e]\!] : \mathbf{list}(\mathcal{D}[\![\tau_1]\!])]\!]$ $\gg\!= \lambda(g,g').\delta_{(\mathbf{match}\ g\ \mathbf{with}\ \mid\text{nil}\to e_1:\tau_2\ \mid h::t\to e_2:\tau_2\ \mathbf{end}:\tau_2,$ $\qquad \mathbf{match}\ g'\ \mathbf{with}\ \mid\text{nil}\to\mathcal{D}[\![e_1:\tau_2]\!]:\mathcal{D}[\![\tau_2]\!]\ \mid h::t\to\mathcal{D}[\![e_2:\tau_2]\!]:\mathcal{D}[\![\tau_2]\!]\ \mathbf{end}:\mathcal{D}[\![\tau_2]\!])}$

Table 4. Small-step coupling semantics (continued). Here, every function of the form $f(g,g') = \lambda(g,g').\delta_{\ldots}$ is extended to $f_\perp$ in the sense of Definition F.29.

Proof. By induction on the type derivation of $e : \tau$. The most interesting case is

$$e : \tau = \mathbf{uniform}(c_1 : \mathbf{float}[B_1; V_1], c_2 : \mathbf{float}[B_2; V_2]) : \mathbf{float}[B; V]\ ,$$

where

(1) $c_1 \leq c_2 c_1 < c_2$,
(2) $B = \{\sim_1 b_1, \ldots, \sim_n b_n\}$,

(3) $V = \top$,
(4) $V_1 = \{v_1, \ldots, v_{n_1}\}$ with recovers(B_1, V_1) and $c_1 \in V_1$ due to has(V_1, c_1) by I.H.
(5) $V_2 = \{w_1, \ldots, w_{n_2}\}$ with recovers(B_2, V_2) and $c_2 \in V_2$ due to has(V_2, c_2) by I.H.

First observe that we have

$$\begin{aligned}
&\mathcal{D}[\![e : \tau]\!] : \mathcal{D}[\![\tau]\!] \\
&= \mathtt{DC}_{B,B_1,B_2,V_1,V_2}(\mathcal{D}[\![c_1 : \mathbf{float}[B_1; V_1]]\!], \mathcal{D}[\![c_2 : \mathbf{float}[B_2; V_2]]\!]) : \mathbf{fin}(|B| + 1) \\
&= \mathtt{DC}_{B,B_1,B_2,V_1,V_2}(k_1 : \mathbf{fin}(|B_1| + 1), k_2 : \mathbf{fin}(|B_2| + 1)) : \mathbf{fin}(|B| + 1)\,,
\end{aligned}$$

where k_i is the unique number such that $c_i \in \text{Intervals}(B_i)_{k_i}$. By Table 1 and Figure 13, this yields

$$\begin{aligned}
&[\![\mathcal{D}[\![e : \tau]\!]]\!] \\
&= [\![\mathbf{discrete}(p_0, \ldots, p_n)]\!] \text{ where } p_k = \text{Uniform}(c_1, c_2)(\text{Intervals}(B)_k)\,.
\end{aligned}$$

Here $c_1 < c_2$, and the uniform CDF is the clamped function

$$F_{c_1,c_2}(x) = \begin{cases} 0 & x \le c_1, \\ \dfrac{x - c_1}{c_2 - c_1} & c_1 < x < c_2, \\ 1 & x \ge c_2. \end{cases}$$

Thus each p_k is the difference of F_{c_1,c_2} at consecutive cut locations (with endpoints $-\infty$ and $+\infty$). Strict and non-strict cuts assign a boundary point to different adjacent intervals, but this does not change these masses because $c_1 < c_2$ makes the uniform distribution atomless.

For Lemma F.43.2, consider the following:

$$\begin{aligned}
&\pi_1([\![e : \tau, \mathcal{D}[\![e]\!] : \mathcal{D}[\![\tau]\!]]\!]) \\
&= [\![e : \tau, \mathcal{D}[\![e]\!] : \mathcal{D}[\![\tau]\!]]\!] \gg\!= \lambda(e_1, e_2).\delta_{e_1} && \text{(definition)} \\
&= (\text{Uniform}(c_1, c_2) \gg\!= \lambda v.\delta_{(v:\mathbf{float}[B;V], v_i:\mathbf{fin}(|B|+1))}) \gg\!= \lambda(e_1, e_2).\delta_{e_1} \\
&\qquad && \text{(Table 3, } v_i \text{ is unique number s.t. } v \in \text{Intervals}(B)_{v_i}) \\
&= \text{Uniform}(c_1, c_2) \gg\!= \lambda v.(\delta_{(v:\mathbf{float}[B;V], v_i:\mathbf{fin}(|B|+1))} \gg\!= \lambda(e_1, e_2).\delta_{e_1}) \\
&\qquad && \text{(monad assoc., } v_i \text{ is unique number s.t. } v \in \text{Intervals}(B)_{v_i}) \\
&= \text{Uniform}(c_1, c_2) \gg\!= \lambda v.\delta_{v:\mathbf{float}[B;V]} \\
&= \text{Uniform}(c_1, c_2) \\
&= [\![e : \tau]\!]\,. && \text{(Table 1)}
\end{aligned}$$

For Lemma F.43.2, consider the following:

$$\begin{aligned}
&\pi_2([\![e : \tau, \mathcal{D}[\![e]\!] : \mathcal{D}[\![\tau]\!]]\!]) \\
&= [\![e : \tau, \mathcal{D}[\![e]\!] : \mathcal{D}[\![\tau]\!]]\!] \gg\!= \lambda(e_1, e_2).\delta_{e_2} && \text{(definition)} \\
&= (\text{Uniform}(c_1, c_2) \gg\!= \lambda v.\delta_{(v:\mathbf{float}[B;V], v_i:\mathbf{fin}(|B|+1))}) \gg\!= \lambda(e_1, e_2).\delta_{e_2} \\
&\qquad && \text{(Table 3, } v_i \text{ is unique number s.t. } v \in \text{Intervals}(B)_{v_i}) \\
&= \text{Uniform}(c_1, c_2) \gg\!= \lambda v.(\delta_{(v:\mathbf{float}[B;V], v_i:\mathbf{fin}(|B|+1))} \gg\!= \lambda(e_1, e_2).\delta_{e_2}) \\
&\qquad && \text{(monad assoc., } v_i \text{ is unique number s.t. } v \in \text{Intervals}(B)_{v_i}) \\
&= \text{Uniform}(c_1, c_2) \gg\!= \lambda v.\delta_{v_i:\mathbf{fin}(|B|+1)} && \text{(monad return, } v_i \text{ is unique number s.t. } v \in \text{Intervals}(B)_{v_i}) \\
&= [\![\mathbf{discrete}(p_0, \ldots, p_n)]\!] \text{ where } p_k = \text{Uniform}(c_1, c_2)(\text{Intervals}(B)_k) && \text{(by construction)} \\
&= [\![\mathcal{D}[\![e]\!] : \mathcal{D}[\![\tau]\!]]\!]\,. && \text{(Table 1)}
\end{aligned}$$

Now, the remaining cases are either (i) trivial base cases or (ii) immediately follow from the I.H. and Lemma F.44 further below. We demonstrate this for the case

$$e : \tau = \mathbf{uniform}(e_1 : \mathbf{float}[B_1; V_1], e_2 : \mathbf{float}[B_2; V_2]) : \mathbf{float}[B; V]\,,$$

where